SOFTWARE DEVELOPER

KARTHICKRAJA.M

Address : 12-1-2d, Krishna Nagar,Nilakottai, Dindigul, Tamilnadu, India

Email : karthickrajam8100@gmail.com , **Mobile No** : 6380528683

SUMMARY

- Embedded Software Developer with Hands-on experience in IoT, microcontrollers, and real-time systems. Skilled in **ESP32, Raspberry Pi**, **LoRa**, **Zigbee**, and **MQTT** with strong knowledge of Python, Embedded C, and cloud integration. Passionate about building innovative IoT solutions for automation, monitoring, and smart systems.

EDUCATION QUALIFICATION

- **Computer Science with Data Analytics**, **Master of Science,** Completed, June 2022. Madurai Kamaraj University, Madurai Marks 82%.

- **Computer Science, Bachelor of Science**, Completed, March 2020. Mary Matha College of Arts & Science, Periyakulam. Marks 67%.

- **12th**, Completed, March 2017. H.N.U.P.R. Matric Hr. Sec. School, Nilakottai. State Board – Marks 65.5%.

- **10th**, Completed, March 2015. H.N.U.P.R. Matric Hr. Sec. School, Nilakottai. State Board – Marks 70%.

WORKING EXPERIENCE

The Petal Automations, Madurai.
Embedded Software Engineer and Python Developer
11-07-2022 – Present

CERTIFICATON COURSES

- Python Programming (Received from SSI)
- Machine Learning
- MyCaptain (Nov 2020) - It was a one month internship work as campus ambassador for my college.
- Hindi

GitHub Link

https://github.com/karthickrajathedeveloper

Python Library Link

https://pypi.org/user/karthickraja.thedeveloper

LANGUAGES

- English
  NativeorBilingualProficiency

- Tamil
  Full Professional Proficiency

1. Smart Gym

Team Size: 2
Role: Embedded Software and Python Developer.
Responsibility: Writing code for controllers such as Arduino Mega, ESP8266 and Raspberry pi.
Technology used: IOT, UART and RS232 to USB Communication
Project Overview: Smart Gym system is used for automatically allocate locker to the gym members and also maintain the locker details and also weighing scale send the weight. And update the locker status and Object status which is inside of the locker to the front end using IOT.

2. IOT LAB KIT V 2 & 2.1

Team Size: 3
Role: Embedded Software Developer.
Responsibility: Writing code in ESP32.
Technology used: IOT, UART, SPI, RS232 to USB, USB to TTL, RFID, I2S, Bluetooth, MQTT and I2C communication Protocols.
Project Overview: It is for student learning purpose. Learn about how to interface I/O and sensors modules with IOT.

3. Ambulance Alert

Team Size: 2.
Role: Embedded Software Developer.
Responsibility: Writing a Code in LoRa.
Technology used: Bluetooth Communication Protocol

Project Overview: Developed a smart medical emergency alert system where crossing an emergency gate triggers notification to hospital staff. Enabled ward assistants to receive real-time alerts and prepare for patient admission.

4. Nurse Calling

System Team Size: 2
Role: Embedded Software Developer
Responsibility: Writing a code in Zigbee.
Technology used: Zigbee, Raspberry pi (Micro python)

Project Overview: A wireless nurse calling system that covers up to 1500 meters inside a hospital using Zigbee communication contains features like code blue alert, nurse call emergency alert, housekeeping call alert, Extra assistance call alert and Toilet and hand unit call alert and nursing station emergency alert, etc.

ENGINEERING / TECHNOLOGY

Languages -
C, C++, Embedded C, Python, Micro-Python

Database -
MYSQL.

Web designing -
HTML.

IDE –
Pycharm, Arduino, VS code.

Controllers –
Nodemcu, ESP32, Arduino UNO and MEGA, Raspberry pi, STC15, PIC16F716.

RF-
WiFi, BLE, LoRa, zigbee,

5. **Steam**

Team Size: 5
Role: Embedded Software Developer
Responsibility: Writing a code in STC15F104AS, STC15W408, ESP32.
Technology used: One-Wire protocol.

Project Overview: Developed an IoT learning kit to interface sensors and controllers, enabling students to understand robotics and embedded systems. Built mini robots through practical application to enhance experiential learning.

6. **Portfolio Monitor**

Team Size: 2
Role: Senior Embedded Software Developer
Responsibility: Developed embedded firmware for ESP32.
Technology used: E-Paper Display, API and MQTT.
Features: Real-time stock updates, RGB + Buzzer alerts, Playlist & Single Mode, BMS integration, OTA updates

Project Overview: Developed a portable stock portfolio monitoring system using ESP32, E-Paper display, REST APIs, and MQTT. The system supports real-time stock updates with visual and auditory alerts, playlist/single display modes, battery management system (BMS) for portable operation, and OTA (Over-the-Air) firmware updates.

ACADEMIC PROJECTS

- Smart Dustbin using **Raspberry pi** - The project is used to automatically open and close the lid and when it's overflows cardinally send the WhatsApp message and normal SMS.
- Smart Garbage Vehicle using **ESP32-CAM** – Built a smart garbage collection vehicle with an **ESP32-CAM** that detects whether a dustbin is full using image processing. If full, the system shares the **GPS location** of the bin and streams **live video** of the vehicle through email while autonomously moving towards the location for collection.

PERSONAL INFORMATION

Father Name	:	Murugan.S
DOB	:	08/01/2000
Gender	:	Male
Marital Status	:	Single
Nationality	:	Indian

DECLARATION

I do hereby declare that the above information is true to the best of my knowledge.

SIGNATURE
M.KARTHICK RAJA